

Azure Data Factory

Building and Parameterizing a Pipeline with Set Variable Activities

This guide walks through creating a Data Factory pipeline (**df3354**) that calculates the area of a rectangle using Set Variable activities. It starts with hardcoded values, then shows how to convert those values into pipeline parameters so the pipeline can be run with different inputs at execution time.

Part 1	Create the pipeline and set up variables with fixed values
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Part 2	Calculate area from the variables and run/debug the pipeline
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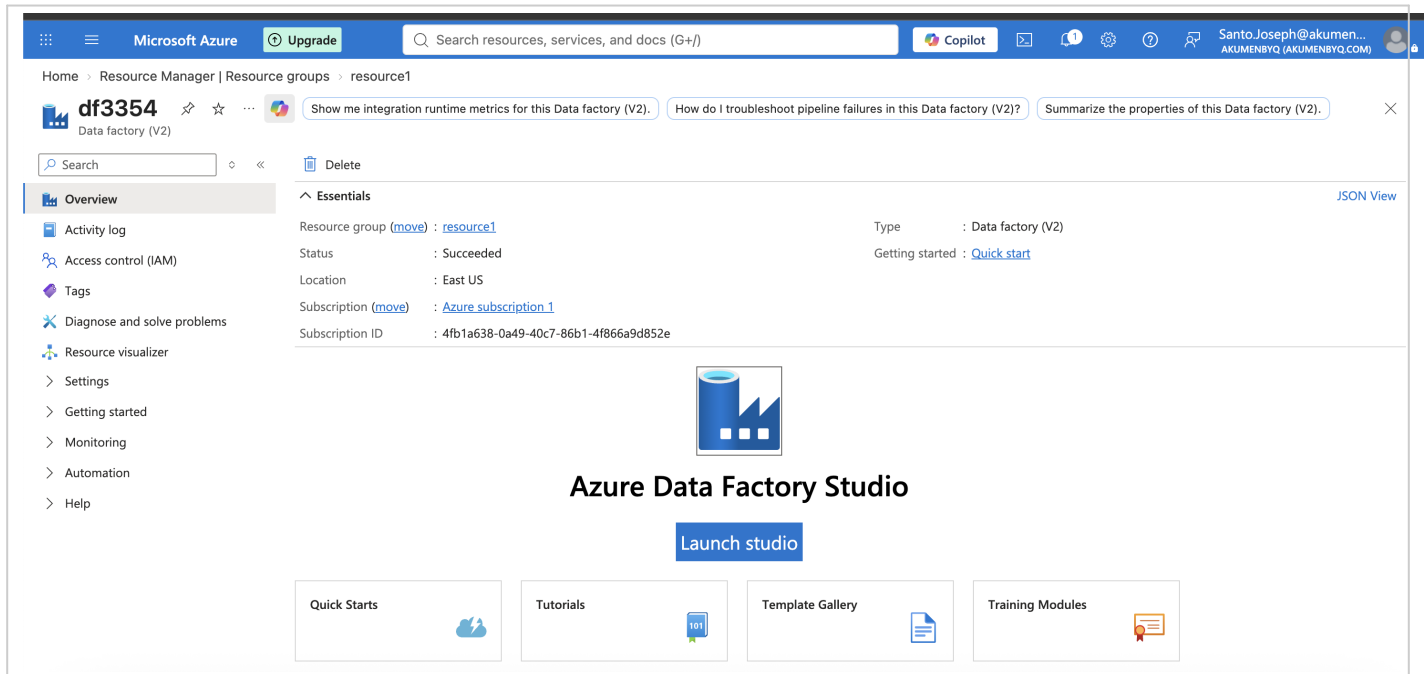
Part 3	Convert hardcoded values into pipeline parameters
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Part 4	Run the pipeline with custom parameter values
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Part 1 — Create the Pipeline and Define Variables

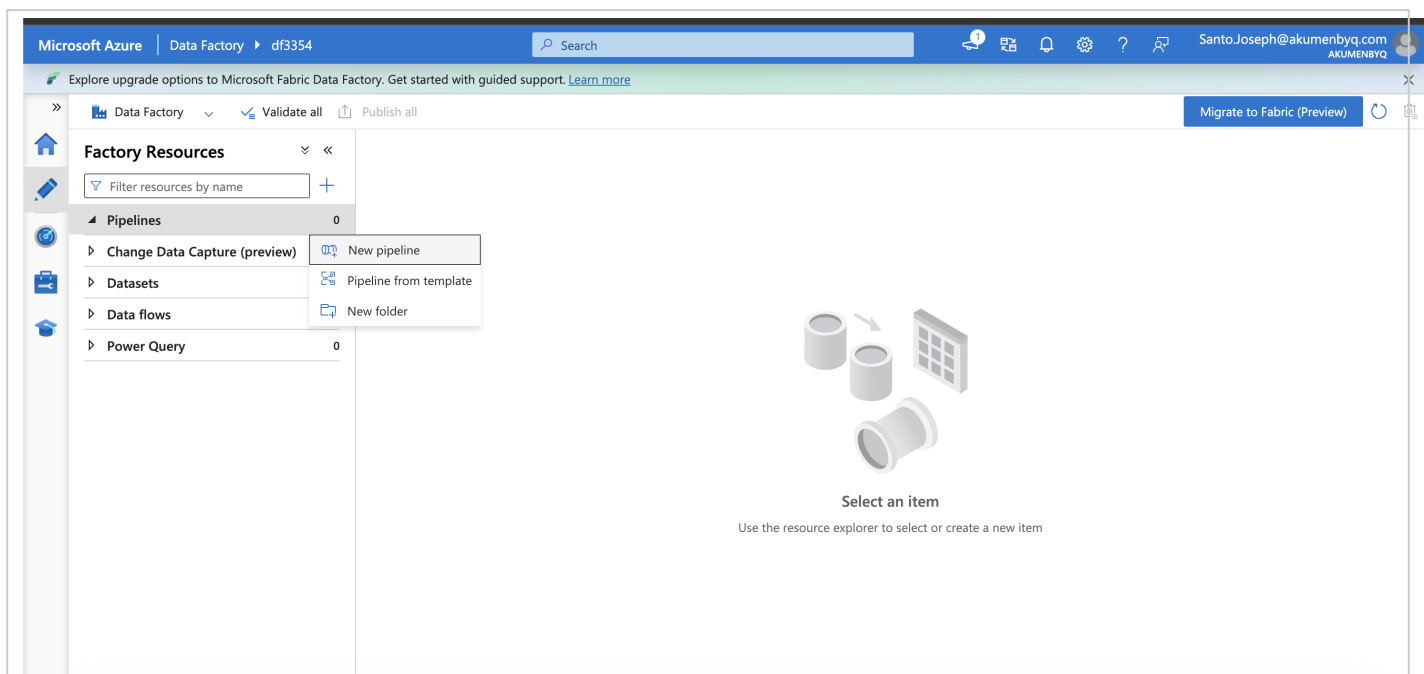
1. Open Azure Data Factory Studio

From the df3354 Data Factory resource in the Azure Portal, select **Launch studio** to open Azure Data Factory Studio, where pipelines are authored.



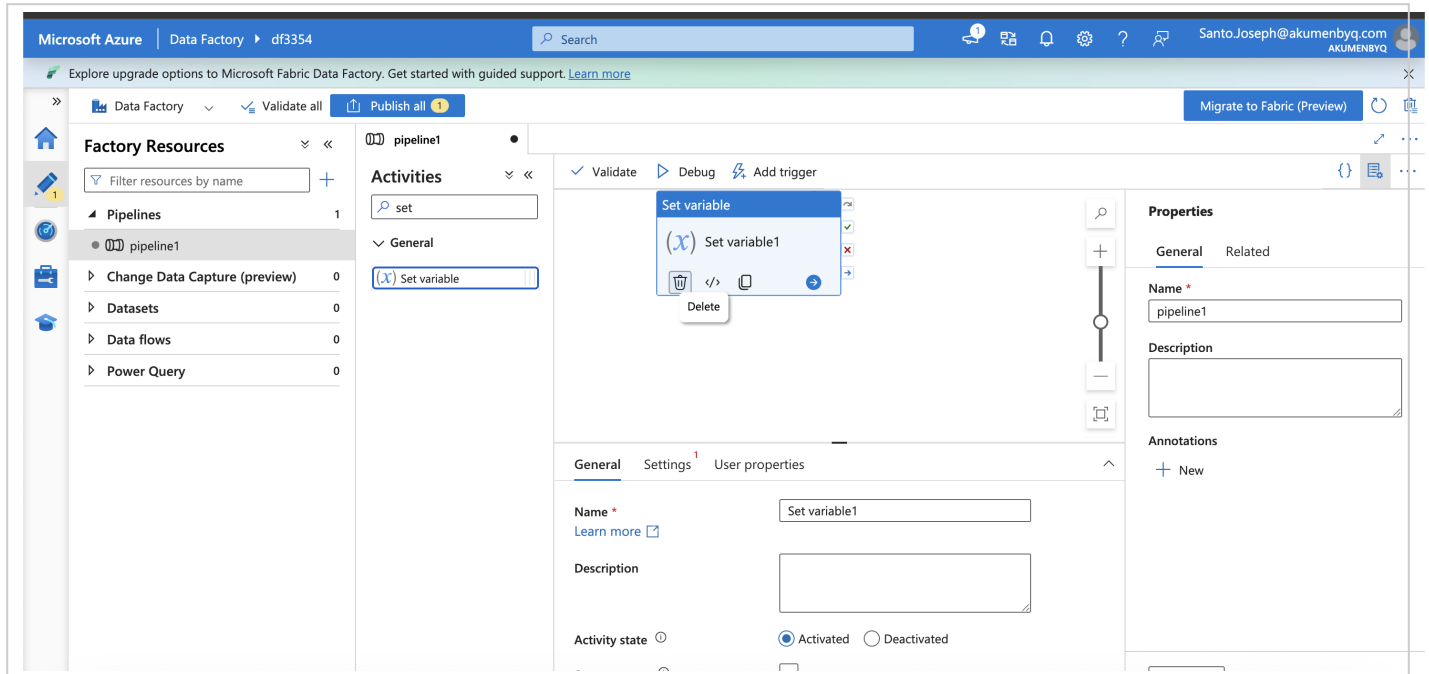
2. Create a New Pipeline

In the Author section, expand **Factory Resources**, click the + icon next to Pipelines, and choose **New pipeline** to create a blank pipeline canvas.



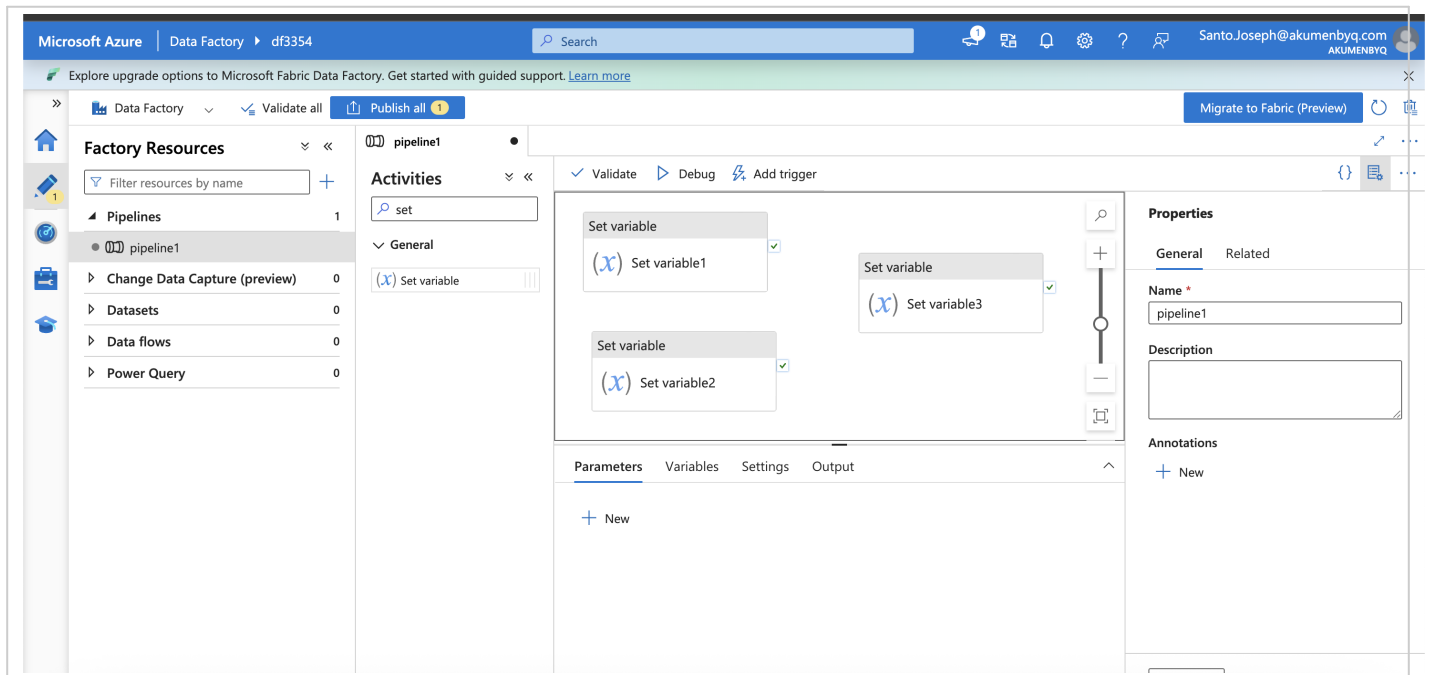
3. Add a Set Variable Activity

Search for **Set variable** in the Activities pane and drag it onto the canvas. The pipeline is renamed to **pipeline1** in the Properties panel on the right.



4. Add Two More Set Variable Activities

Two additional Set variable activities are added to the canvas: **Set variable1**, **Set variable2**, and **Set variable3**. These will hold the length, breadth, and calculated area values.



5. Configure the "length" Variable

Select the first Set variable activity, open the **Settings** tab, set **Name** to **length**, and set **Value** to **10**.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' with 'pipeline1' selected. The 'Activities' pane shows a 'Set variable' activity. The main canvas displays a pipeline diagram with two 'Set variable' activities: one for 'length' and one for 'breadth'. The 'Settings' tab is active for the 'length' activity, showing 'Variable type' as 'Pipeline variable', 'Name' as 'length', and 'Value' as '10'. The 'Properties' pane on the right shows the 'General' tab with 'Name' as 'pipeline1'.

6. Configure the "breadth" Variable

Select the second Set variable activity, set **Name** to **breadth**, and set **Value** to **16**.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' with 'pipeline1' selected. The 'Activities' pane shows a 'Set variable' activity. The main canvas displays a pipeline diagram with two 'Set variable' activities: one for 'length' and one for 'breadth'. The 'Settings' tab is active for the 'breadth' activity, showing 'Variable type' as 'Pipeline variable', 'Name' as 'breadth', and 'Value' as '16'. The 'Properties' pane on the right shows the 'General' tab with 'Name' as 'pipeline1'.

Part 2 — Calculate the Area and Run the Pipeline

7. Calculate "area" from the Two Variables

On the third Set variable activity, set **Name** to **area** and set **Value** to the expression `@mul(variables('length'),variables('breadth'))`, connecting both prior activities to this one on the canvas.

The screenshot shows the Microsoft Azure Data Factory pipeline editor for a pipeline named 'pipeline1'. The canvas displays three 'Set variable' activities: 'length', 'breadth', and 'area'. The 'area' activity is selected, and its properties are shown in the right-hand pane. The 'Name' is set to 'area', and the 'Value' is set to the expression `@mul(variables('length'),variables('breadth'))`. The 'Variable type' is set to 'Pipeline variable'.

Factory Resources

- Pipelines: 1
 - pipeline1
- Datasets: 0
- Data flows: 0
- Power Query: 0

Activities

- set
- Set variable

Properties

- Name: pipeline1
- Description:
- Annotations: + New

Settings

- Variable type: Pipeline variable
- Name: area
- Value: `@mul(variables('length'),variables('breadth'))`

8. Debug the Pipeline

Click **Debug** to run the pipeline. The **Output** tab lists each activity — **length**, **breadth**, and **area** — all showing a **Succeeded** status.

The screenshot shows the Microsoft Azure Data Factory pipeline editor for the same pipeline 'pipeline1'. The 'Output' tab is selected, displaying a table of activity results. All three activities ('length', 'breadth', and 'area') are shown with a 'Succeeded' status.

Factory Resources

- Pipelines: 1
 - pipeline1
- Datasets: 0
- Data flows: 0
- Power Query: 0

Activities

- set
- Set variable

Output

Activity name	Activity st...	Activit...	Run start
area	✓ Succeeded	Set variable	8/18/2026, 1:05:50 PM
length	✓ Succeeded	Set variable	8/18/2026, 1:05:49 PM
breadth	✓ Succeeded	Set variable	8/18/2026, 1:05:49 PM

9. Review the Output Value

Clicking the output icon for the **area** activity shows the calculated result: `{"name": "area", "value": 160}`, confirming $10 \times 16 = 160$.

The screenshot displays the Microsoft Azure Data Factory portal. The left sidebar shows 'Factory Resources' with a list of pipelines, including 'pipeline1'. The main workspace shows the 'Activities' section for 'pipeline1', with a 'Set variable' activity selected. The 'Output' tab is active, showing the calculated result for the 'area' activity: `{"name": "area", "value": 160}`. The 'Properties' panel on the right shows the 'General' tab with the 'Name' field set to 'pipeline1'. The 'Run start' button is visible at the bottom right.

Activity	Status	Set variable	Run start
area	Succeeded	Set variable	8/18/2026, 1:05:50 PM
length	Succeeded	Set variable	8/18/2026, 1:05:49 PM
breadth	Succeeded	Set variable	8/18/2026, 1:05:49 PM

Part 3 — Convert Hardcoded Values into Pipeline Parameters

10. Open the Pipeline Expression Builder

To make the pipeline reusable, the fixed values are replaced with pipeline parameters. Opening the expression builder for **breadth** and typing `@pipeline().parameters.breadth` initially returns an error because the referenced parameter does not yet exist on the pipeline.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of resources including 'pipeline1'. The main workspace displays the 'Activities' pane with a 'Set variable' activity. The 'Settings' tab for this activity is active, showing the 'Variable type' as 'Pipe' and the 'Name' as 'length'. The 'Value' field is empty. The 'Pipeline expression builder' pane on the right shows the expression `@pipeline().parameters.breadthpipeline().parameters.lengthpipeline().parameters.breadth`. A red error message indicates that the parameter `breadthpipeline()` was not found under `pipeline1`. The 'Parameters' tab is active, showing a list of parameters: `breadth` and `length`.

11. Correct the Parameter Reference

A further attempt to reference `lengthpipeline()` also fails validation, since `pipeline()` is a fixed system function and parameter names must be appended after `.parameters`.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of resources including 'pipeline1'. The main workspace displays the 'Activities' pane with a 'Set variable' activity. The 'Settings' tab for this activity is active, showing the 'Variable type' as 'Pipe' and the 'Name' as 'length'. The 'Value' field is empty. The 'Pipeline expression builder' pane on the right shows the expression `@pipeline().parameters.lengthpipeline().parameters.breadthpipeline().parameters.lengthpipeline().parameters.breadth`. A red error message indicates that the parameter `lengthpipeline()` was not found under `pipeline1`. The 'Parameters' tab is active, showing a list of parameters: `breadth` and `length`.

12. Set the Correct Expression for "breadth"

The corrected expression `@pipeline().parameters.breadth` is entered and validates successfully, with **breadth** and **length** now listed as available parameters.

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'pipeline1'. The main workspace shows the 'Activities' tab for 'pipeline1', with a 'Set variable' activity selected. The 'Settings' tab for this activity is open, showing the 'Variable type' as 'Pipe' and the 'Name' as 'breadth'. The 'Value' field is empty. On the right, the 'Pipeline expression builder' is open, showing the expression `@pipeline().parameters.breadth` in the text area. Below the text area, the 'Parameters' tab is selected, displaying a search bar and a list of available parameters: 'breadth' and 'length'.

13. Update the "area" Expression to Use Parameters

The **area** activity's value is updated to `@mul(pipeline().parameters.length, pipeline().parameters.breadth)`, so the calculation now pulls from pipeline parameters instead of fixed variables.

The screenshot displays the Microsoft Azure Data Factory portal for a specific Data Factory instance (df3354). The interface is divided into several sections:

- Factory Resources:** A sidebar on the left showing a list of resources including Pipelines (1), Change Data Capture (preview) (0), Datasets (0), Data flows (0), and Power Query (0).
- Activities:** A central pane showing the 'pipeline1' activity. Under the 'General' tab, a 'Set variable' activity is listed.
- Diagram:** A visual representation of the pipeline logic. It shows three 'Set variable' activities: 'length', 'breadth', and 'area'. Arrows indicate that 'length' and 'breadth' are inputs to the 'area' activity.
- Properties:** A panel on the right showing the configuration for the selected 'Set variable' activity. The 'Name' is 'area'. The 'Value' is set to the expression `@mul(pipeline().parameters.length, pipeline().parameters.breadth)`.
- Settings:** A tab at the bottom of the Properties panel showing the 'Variable type' as 'Pipeline variable'.

Part 4 — Run the Pipeline with Custom Parameter Values

14. Trigger a Pipeline Run

Running the pipeline now prompts for the **Pipeline run** parameters. Both **length** and **breadth** are marked **Required** and must be filled in before the run can start.

Microsoft Azure | Data Factory | df3354

Explore upgrade options to Microsoft Fabric Data Factory. Get started with guided support. [Learn more](#)

Factory Resources

- Pipelines: 1
 - pipeline1
 - Change Data Capture (preview): 0
 - Datasets: 0
 - Data flows: 0
 - Power Query: 0

Activities

- set
 - General
 - (x) Set variable

Set variable (x) length

Set variable (x) breadth

Parameters Variables Settings

All status

Showing 1 - 3 of 3 items

Activity name	Activity
area	✓ Succ
length	✓ Succ
breadth	✓ Succ

Pipeline run

Parameters

Name	Type	Value
length	Int	Value Required
breadth	Int	Value Required

15. Enter New Parameter Values

New values are entered for this run: **length = 15** and **breadth = 20**. Clicking **OK** starts the pipeline with these inputs.

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Factory Resources

- Pipelines: 1
 - pipeline1
 - Change Data Capture (preview): 0
 - Datasets: 0
 - Data flows: 0
 - Power Query: 0

Activities

- set
 - General
 - (x) Set variable

Set variable (x) length

Set variable (x) breadth

Parameters Variables Settings

All status

Showing 1 - 3 of 3 items

Activity name	Activity
area	✓ Succ
length	✓ Succ
breadth	✓ Succ

Pipeline run

Parameters

Name	Type	Value
length	Int	15
breadth	Int	20

16. View the Final Calculated Output

The **area** activity output now returns **{"name": "area", "value": 300}**, confirming the pipeline correctly calculated $15 \times 20 = 300$ using the new parameter values — proving the pipeline is now fully parameterized and reusable.

The screenshot displays the Microsoft Azure Data Factory interface for a pipeline named 'pipeline1'. The 'Output' window shows the final calculated output as a JSON object: `{ "name": "area", "value": 300 }`. The 'Properties' pane on the right indicates the pipeline name is 'pipeline1'. The bottom table provides a summary of the pipeline's execution:

Activity Name	Status	Activity Type	Execution Time
area	Succeeded	Set variable	8/18/2026, 1:10:29 PM
breadth	Succeeded	Set variable	8/18/2026, 1:10:28 PM
length	Succeeded	Set variable	8/18/2026, 1:10:28 PM